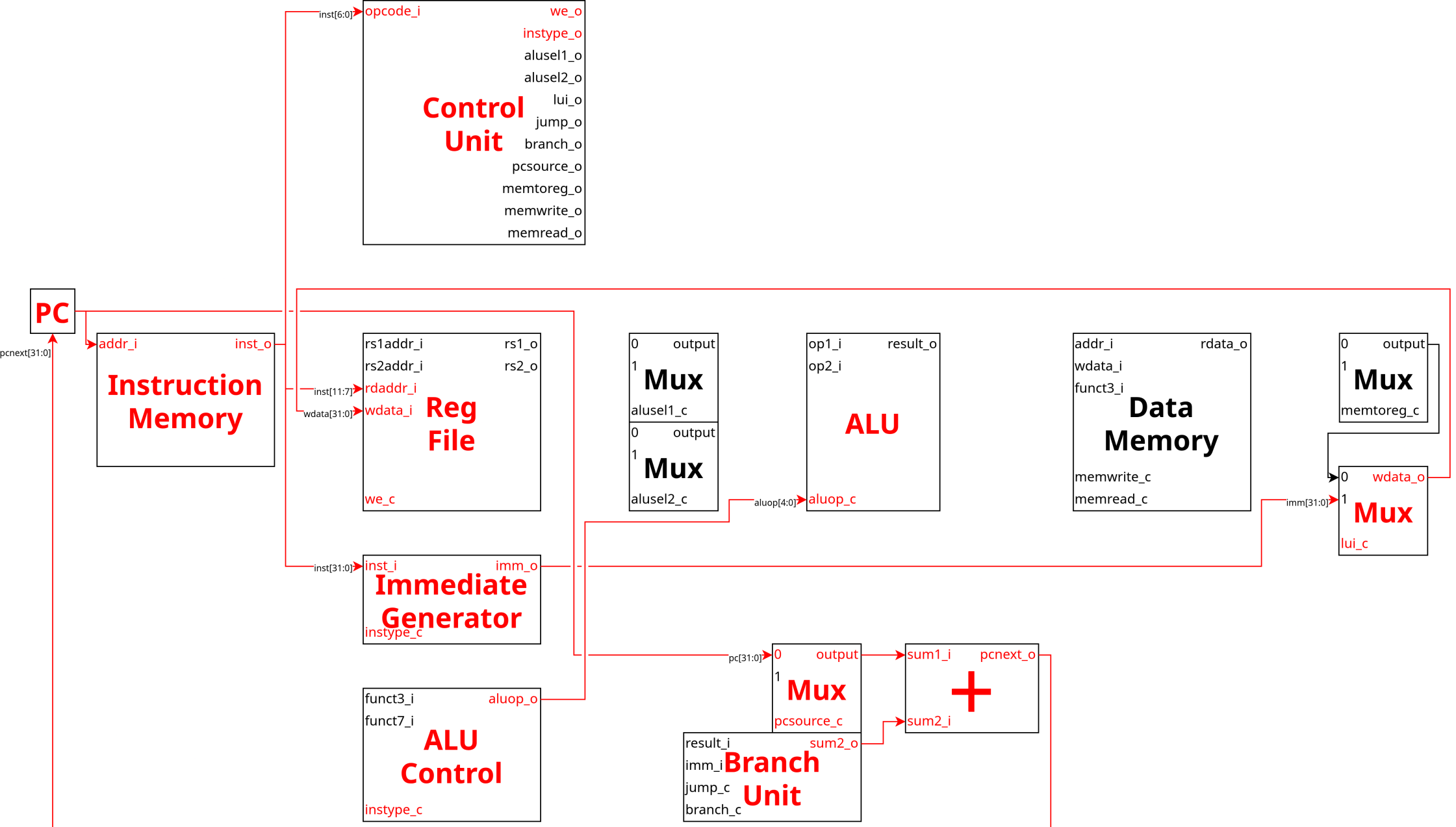
inst[6:0]
opcode_i
we_o
instype_o
alusel1_o
alusel2_o
lui_o
jump_o
branch_o
pcsource_o
memtoreg_o
memwrite_o
memread_o
Control Unit
PC
pcnext[31:0]
addr_i
inst_o
Instruction Memory
inst[11:7]
wdata[31:0]
rs1addr_i
rs2addr_i
rdaddr_i
wdata_i
we_c
rs1_o
rs2_o
Reg File
0
1
output
Mux
alusel1_c
alusel2_c
aluop[4:0]
op1_i
op2_i
aluop_c
result_o
ALU
addr_i
wdata_i
funct3_i
memwrite_c
memread_c
rdata_o
Data Memory
memtoreg_c
imm[31:0]
wdata_o
lui_c
inst[31:0]
inst_i
imm_o
instype_c
Immediate Generator
funct3_i
funct7_i
aluop_o
instype_c
ALU Control
pc[31:0]
pcsource_c
sum1_i
sum2_i
pcnext_o
+
result_i
imm_i
jump_c
branch_c
sum2_o
Branch Unit